



Alloy Lab Database

A personal research infrastructure for experimental alloy development

Stage 1 — Complete

From a messy desktop folder to a queryable lab notebook



Why This Project

The problem

- Experimental results scattered across spreadsheets and raw instrument files
- No structured way to cross-check a new composition against existing science
- Years of accumulated data in a genuinely messy desktop folder
- No durable, queryable record to eventually feed machine learning



Motive 1 — Practical

A real lab database — one place to record what was made, how, and what it measured; usable later as ML training data.



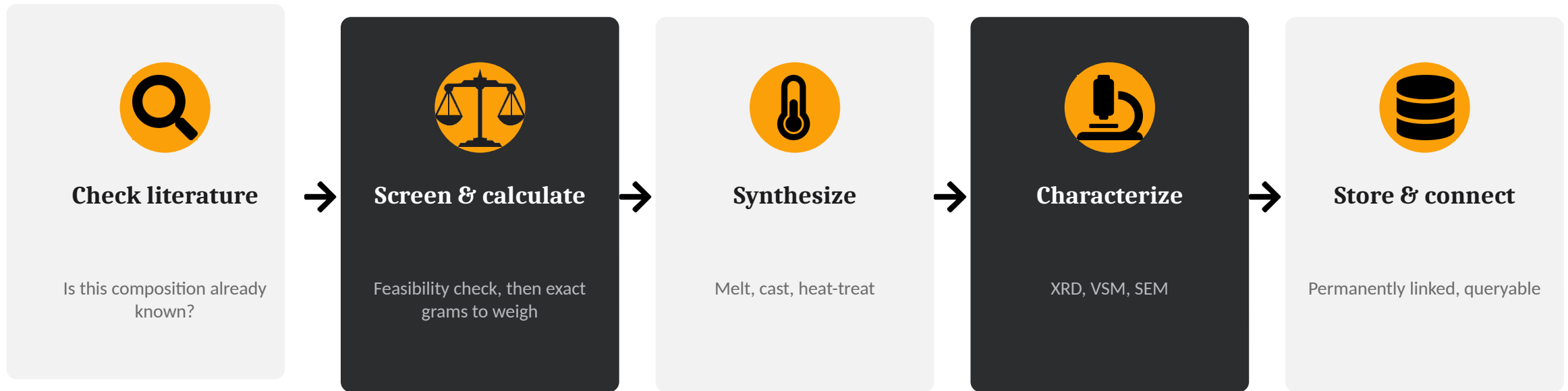
Motive 2 — Personal

Plain curiosity about building this solo — and finally taming a desktop folder years in the making.



The Researcher's Routine

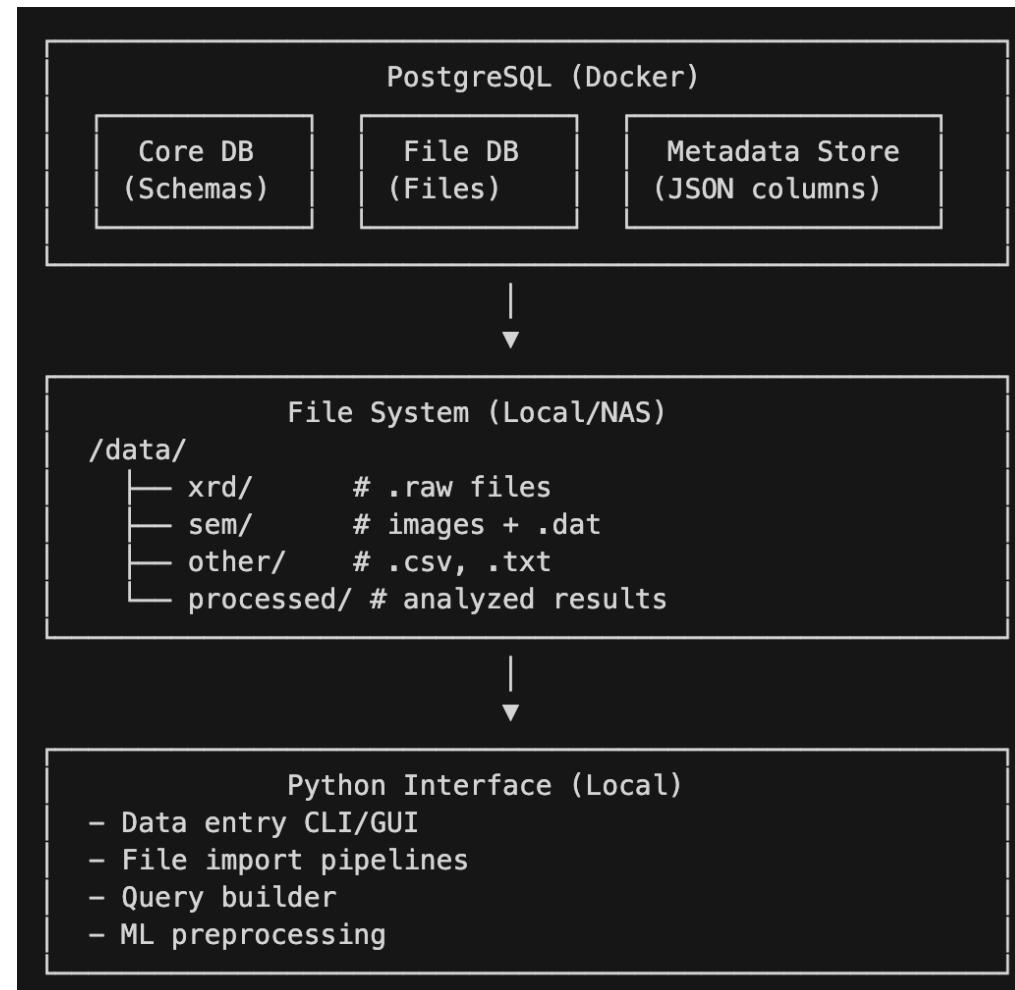
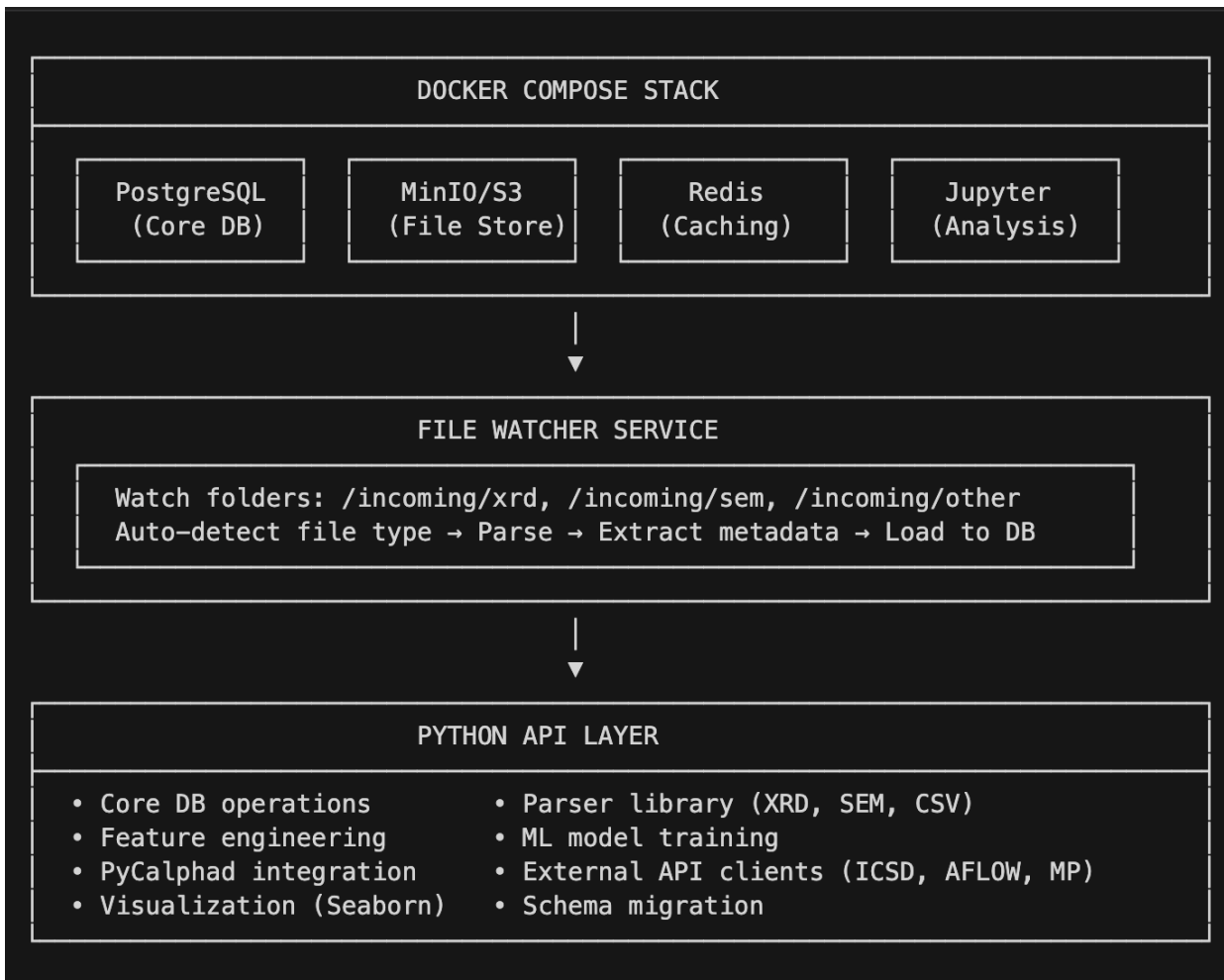
The system was built to directly support this routine — not to replace it.





What Was Planed & What was Built

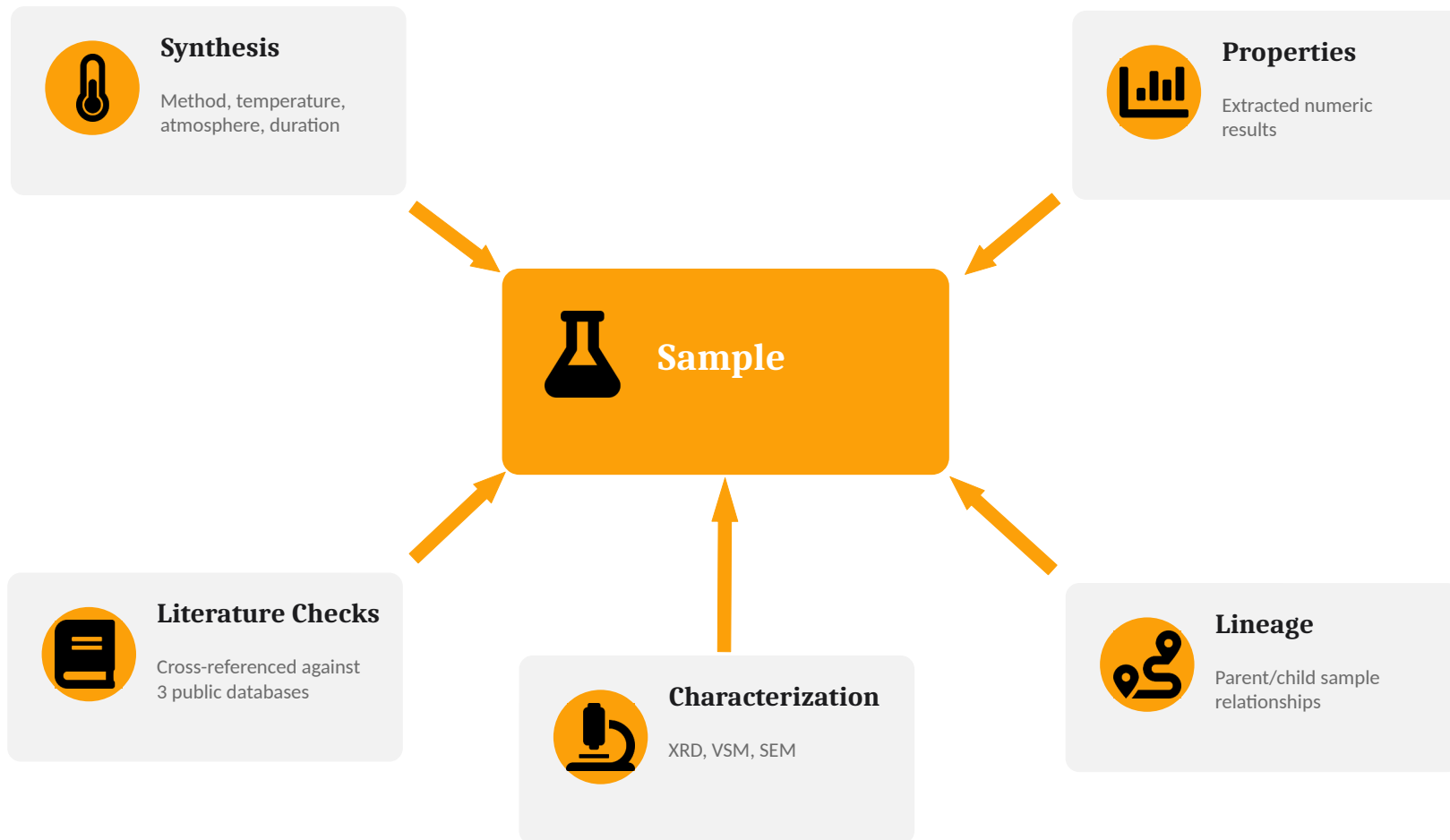
Every sample's full story stays connected — composition, how it was made, and what it measured.





What Was Built

Every sample's full story stays connected — composition, how it was made, and what it measured.



```
alloy_lab
├── samples
│   ├── sample_id (TEXT, UNIQUE)
│   ├── material_class_id (FK → material_classes)
│   ├── composition (JSONB) # atomic fractions
│   ├── parent_sample_id (FK → samples) # family tree
│   ├── mass_grams (REAL)
│   ├── source_type (TEXT) # 'experimental', 'literature'
│   ├── notes (TEXT)
│   └── created_at (TIMESTAMP)
├── compositions
│   ├── sample_id (FK → samples)
│   ├── element (TEXT)
│   ├── weight_percent (REAL)
│   └── atomic_percent (REAL)
├── characterization
│   ├── sample_id (FK → samples)
│   ├── char_type (TEXT) # XRD, SEM, EDS, etc.
│   ├── instrument (TEXT)
│   ├── file_path (TEXT)
│   └── created_at (TIMESTAMP)
├── properties
│   ├── characterization_id (FK → characterization)
│   ├── property_name (TEXT)
│   ├── property_value (REAL)
│   ├── property_unit (TEXT)
│   └── confidence_score (REAL)
├── synthesis
│   ├── sample_id (FK → samples)
│   ├── method (TEXT)
│   ├── temperature (REAL)
│   ├── atmosphere (TEXT)
│   ├── duration_minutes (REAL)
│   └── success (BOOLEAN)
├── literature_sources
│   ├── doi (TEXT, UNIQUE)
│   ├── title (TEXT)
│   ├── authors (TEXT[])
│   └── year (INTEGER)
└── material_classes
    ├── class_name (TEXT, UNIQUE)
    └── description (TEXT)
```

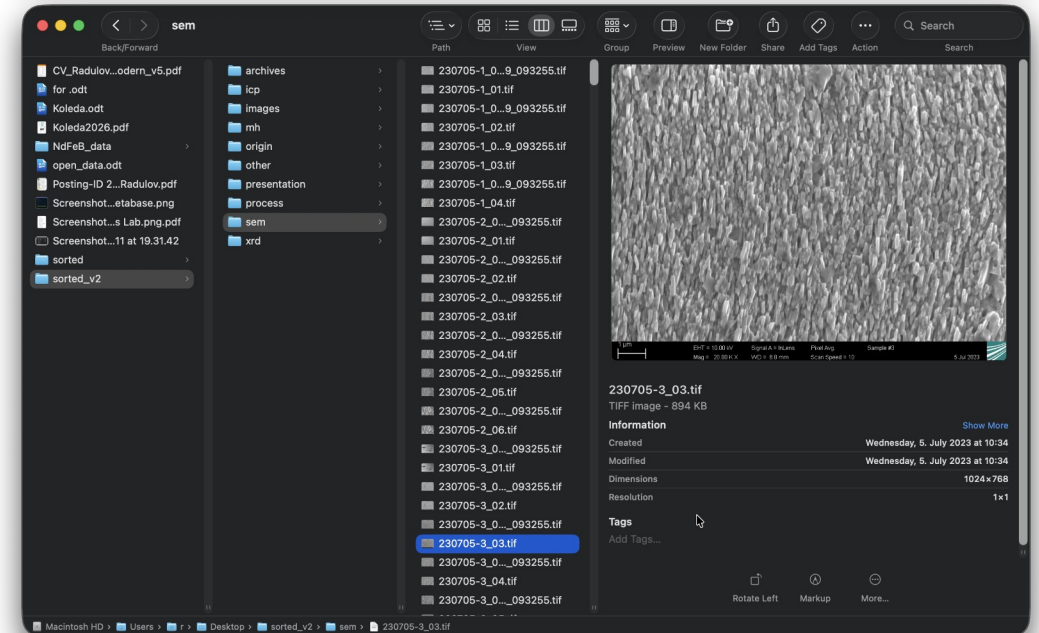
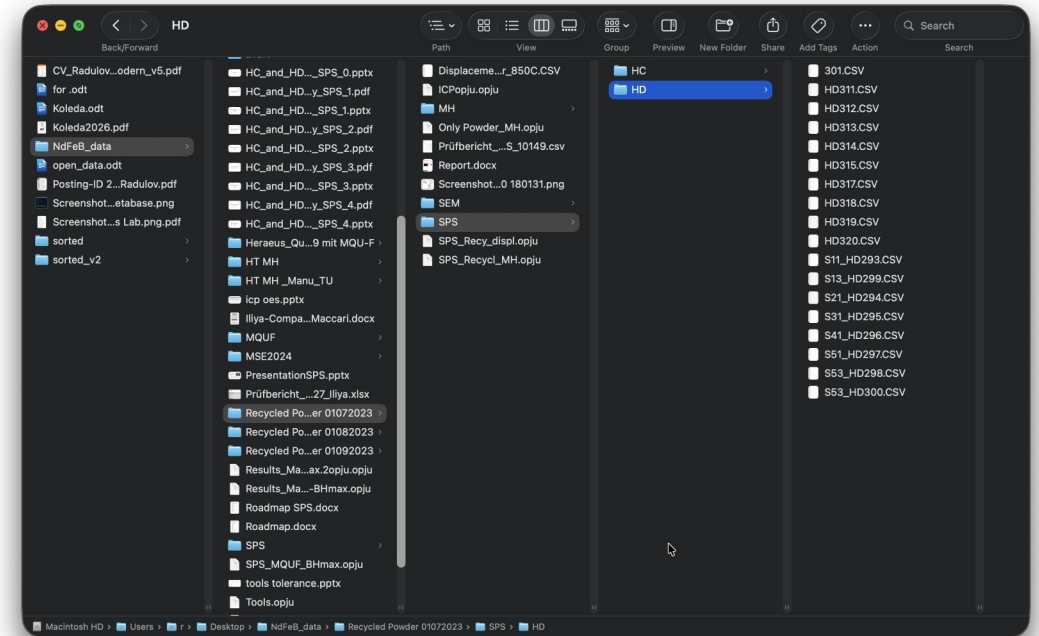


Taming the Data

Years of instrument output, spreadsheets, and project files — sorted automatically.

A standalone utility scans a folder and routes each file into the right category:

- XRD, SEM, magnetic (VSM), ICP composition data, Origin projects, presentations
- Detection by file extension, folder name, and — for ambiguous spreadsheets — file content
- Also attempts to recognize the sample ID directly from the filename

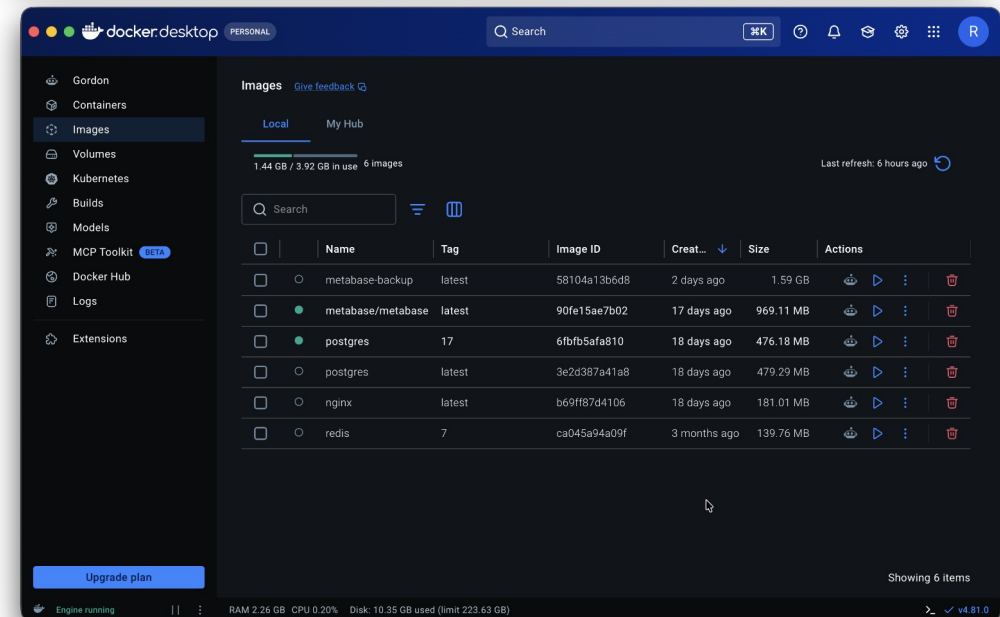
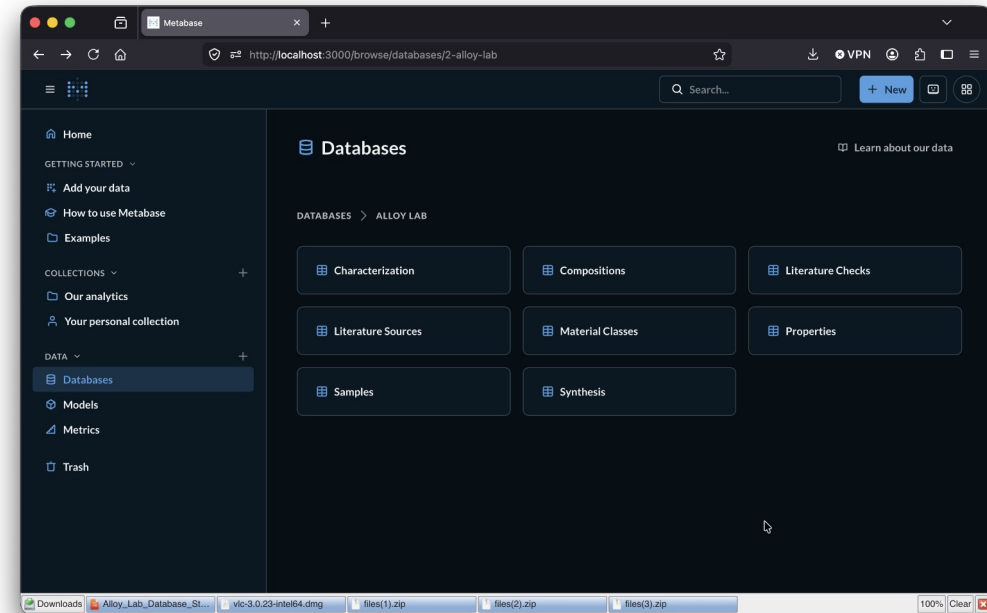




The Database Backbone

A real relational database, not spreadsheets.

- PostgreSQL, running in a self-contained Docker environment
- Every sample's composition, synthesis record, and measured properties stay linked together
- Credentials kept out of source code from day one
- Browsable through Metabase — dashboards without writing queries



Screenshot: Docker containers running



From Formula to Grams

Type a formula. Get exactly what to weigh.

- Accepts natural formula notation (e.g. Fe₆₅Nd₃₀Co₅), atomic % or weight %
- Supports pre-alloys (master alloys) as building blocks, not just pure elements
- Compensates for evaporation losses during melting with an adjustable excess %
- Validates formulas strictly — catches genuinely ambiguous input before it becomes a mistake

Fe₆₅Nd₃₀Co₅ → 10 g target



Fe 4.40 g

Nd 5.24 g

Co 0.36 g



Command-Line Workflow

One command runs the entire entry pipeline.

- Parse the formula
- Run the feasibility screen
- Cross-check the literature
- Calculate exact masses
- Confirm, then store — permanently, and queryable

```
my_db_project — r@Iliyas-MacBook-Air — ..my_db_p...

Alloy Entry Tool

Enter alloy formula (e.g., Fe65Nd30Co5): LaFe11.6Si1.4
Is this at% or wt%? [at%]: at%

Parsing LaFe11.6Si1.4 as at%...
Atomic %: La=82.86, Fe=10.00, Si=7.14

Target total mass in grams [10.0]: 10

Available material classes:
1. Permanent Magnet
2. Soft Magnetic
3. High Entropy Alloy
4. Heusler
5. Single Crystal
Select class [1]: 2

Sample ID [NEW-20260706-001]: LaFeSi-TEST-001

Optional: Add excess for evaporation loss
Enter element:excess% (e.g., Co:5), or press Enter to skip
Excess (e.g., Co:5): la:3

Calculating masses...

Calculation Results

Effective molar mass: 122.683 g/mol
Total target mass: 10.0000 g

Element      at%      wt% target(g)  weigh(g)
-----
La           82.86     93.81     9.3813     9.3813
Fe           10.00      4.55      0.4552     0.4552
Si            7.14      1.64      0.1635     0.1635

Add this alloy to database? [y/N]: y

Adding to database...
Connected to database: alloy_lab
Added sample: LaFeSi-TEST-001 (ID: 11)

Alloy added successfully!
Sample ID: LaFeSi-TEST-001
Mass: 10.0g
Class: Soft Magnetic
Composition: La=82.86 at%, Fe=10.00 at%, Si=7.14 at%
(venv) + my_db_project
```

Screenshot: CLI running the full workflow



Command-Line Workflow

```
my_db_project - r@Ilyas-MacBook-Air - ..my_db_project - -zsh - 81x66

Enter alloy formula (e.g., Fe65Nd30Co5): Fe65Nd30Co5
Is this at% or wt%? [at%]: at%

Parsing Fe65Nd30Co5 as at%...
Atomic %: Fe=65.00, Nd=30.00, Co=5.00

Checking Materials Project...
Retrieving SummaryDoc documents: 100% [ ] 2/2 [00:00<00:00, 15887.52
Target composition: {'Fe': 0.65, 'Nd': 0.3, 'Co': 0.05}

--- Tier 2: similar (same elements, different ratio) ---
mp-1190182 Nd2Fe15Co2 0.070 eV/atom above hull (experimentally known),
distance=0.195
mp-1220336 Nd2Fe4Co13 0.005 eV/atom above hull (theoretical only), dist
ance=0.634

Best match: Nd2Fe15Co2 (Tier 2)
This composition is 0.070 eV/atom above hull

Target total mass in grams [10.0]: 10

Available material classes:
1. Permanent Magnet
2. Soft Magnetic
3. High Entropy Alloy
4. Heusler
5. Single Crystal
Select class [1]: 1

Sample ID [NEW-20260706-001]: NdFeB-20260706-001

Optional: Add excess for evaporation loss
Enter element:excess% (e.g., Co:5), or press Enter to skip
Excess (e.g., Co:5): Nd:3

Calculating masses...

Calculation Results

Effective molar mass: 82.519 g/mol
Total target mass: 10.0000 g

Element at% wt% target(g) weigh(g)
Fe 65.00 43.99 4.3989 4.3989
Nd 30.00 52.44 5.2440 5.4013
Co 5.00 3.57 0.3571 0.3571

Add this alloy to database? [y/N]: y

Adding to database...
Connected to database: alloy_lab
Added sample: NdFeB-20260706-001 (ID: 12)

Alloy added successfully!
Sample ID: NdFeB-20260706-001
Mass: 10.0g
Class: Permanent Magnet
Composition: Fe=65.00 at%, Nd=30.00 at%, Co=5.00 at%
(venv) -> my_db_project
```

```
my_db_project - r@Ilyas-MacBook-Air - ..my_db_project - -zsh - 92x...

ENDOFFILE
(venv) -> my_db_project python alloy_entry_full.py

Alloy Entry Tool (Full Version)
Includes: Calculator | MP Lookup | VEC/5/ΔH_mix Screening

Enter alloy formula (e.g., Fe65Nd30Co5): Fe65Nd30Co5B1
Is this at% or wt%? [at%]: at%

Parsing Fe65Nd30Co5B1 as at%...
Atomic %: Fe=64.36, Nd=29.70, Co=4.95, B=0.99

Running VEC/5/ΔH_mix screening...

Screening Interpretation:
VEC = 6.51 -> Likely BCC solid solution
δ = 0.184 -> Small atomic mismatch: solid solution likely
ΔH_mix = -2.7 kJ/mol -> Weak compound formation

Checking Materials Project...
Retrieving SummaryDoc documents: 100% [ ] 1/1 [00:00<00:00, 6374.321t/s]
Target composition: {'Fe': 0.6435643564356436, 'Nd': 0.297029702970297, 'Co': 0.04950495049504951, 'B': 0.009900990099009901}

--- Tier 2: similar (same elements, different ratio) ---
mp-1192784 NdFeCo3B 0.001 eV/atom above hull (experimentally known), distance=0.607

Target total mass in grams [10.0]: 10

Available material classes:
1. Permanent Magnet
2. Soft Magnetic
3. High Entropy Alloy
4. Heusler
5. Single Crystal
Select class [1]: 1

Sample ID [NEW-20260706-001]: NdFeB-20260706-002

Optional: Add excess for evaporation loss
Excess (e.g., Co:5): Nd:2

Calculating masses...

Calculation Results

Effective molar mass: 81.809 g/mol
Total target mass: 10.0000 g

Element at% wt% target(g) weigh(g)
Fe 64.36 43.93 4.3932 4.3932
Nd 29.70 52.37 5.2371 5.3419
Co 4.95 3.57 0.3566 0.3566
B 0.99 0.13 0.0131 0.0131

Add this alloy to database? [y/N]: y

Adding to database...
Connected to database: alloy_lab
Added sample: NdFeB-20260706-002 (ID: 13)

Alloy added successfully!
Sample ID: NdFeB-20260706-002
Mass: 10.0g
Class: Permanent Magnet
VEC: 6.51
δ: 0.184
(venv) -> my_db_project
```

```
my_db_project - r@Ilyas-MacBook-Air - ..my_db_project - -zsh - 110...

(venv) -> my_db_project git:(master) x python alloy_entry_full.py

Alloy Entry Tool (Full Version)
Includes: Calculator | MP Lookup | VEC/5/ΔH_mix Screening

Enter alloy formula (e.g., Fe65Nd30Co5): Fe65Nd30Co5
Is this at% or wt%? [at%]: at%

Parsing Fe65Nd30Co5 as at%...
Atomic %: Fe=65.00, Nd=30.00, Co=5.00

Running VEC/5/ΔH_mix screening...

Screening Interpretation:
VEC = 6.55 -> Likely BCC solid solution
δ = 0.180 -> Small atomic mismatch: solid solution likely
ΔH_mix = -2.6 kJ/mol -> Weak compound formation

Checking Materials Project...
Retrieving SummaryDoc documents: 100% [ ] 2/2 [00:00<00:00, 2421
4.81it/s]
Target composition: {'Fe': 0.65, 'Nd': 0.3, 'Co': 0.05}

--- Tier 2: similar (same elements, different ratio) ---
mp-1190182 Nd2Fe15Co2 0.070 eV/atom above hull (experimentally known), distance=0.195
mp-1220336 Nd2Fe4Co13 0.005 eV/atom above hull (theoretical only), distance=0.634

Checking QCMD...
Target composition: {'Fe': 0.65, 'Nd': 0.3, 'Co': 0.05}

--- Tier 2: similar (same elements, different ratio) ---
qcmd-1782404 Co2Fe15Nd2 0.172 eV/atom above hull (computed/hypothetical only), distance=0.123
qcmd-29251 Co2Fe15Nd2 0.067 eV/atom above hull (experimentally known (ICSD)), distance=0.195
qcmd-525591 CoFe2Nd 0.484 eV/atom above hull (computed/hypothetical only), distance=0.200
qcmd-861437 CoFeNd 0.485 eV/atom above hull (computed/hypothetical only), distance=0.317
qcmd-964689 CoFeNd 0.574 eV/atom above hull (computed/hypothetical only), distance=0.357
qcmd-967164 CoFeNd 1.224 eV/atom above hull (computed/hypothetical only), distance=0.317
qcmd-536856 CoFeNd2 0.338 eV/atom above hull (computed/hypothetical only), distance=0.400
qcmd-414004 Co2FeNd 0.387 eV/atom above hull (computed/hypothetical only), distance=0.450

Target total mass in grams [10.0]: 10

Available material classes:
1. Permanent Magnet
2. Soft Magnetic
3. High Entropy Alloy
4. Heusler
5. Single Crystal
Select class [1]: 1

Sample ID [NEW-20260707-001]: Fe65Nd30Co5-20260707-001

Optional: Add excess for evaporation loss
Excess (e.g., Co:5): Nd:2

Calculating masses...

Calculation Results

Effective molar mass: 82.519 g/mol
Total target mass: 10.0000 g

Element at% wt% target(g) weigh(g)
Fe 65.00 43.99 4.3989 4.3989
Nd 30.00 52.44 5.2440 5.3409
Co 5.00 3.57 0.3571 0.3571

Add this alloy to database? [y/N]: y

Adding to database...
Connected to database: alloy_lab
Added sample: Fe65Nd30Co5-20260707-001 (ID: 16)

Alloy added successfully!
Sample ID: Fe65Nd30Co5-20260707-001
Mass: 10.0g
Class: Permanent Magnet
VEC: 6.55
δ: 0.180
ΔH_mix: -2.55 kJ/mol

Storing literature check results...
Logged materials_project tier 2: Nd2Fe15Co2
Logged materials_project tier 2: Nd2Fe4Co13
Logged qcmd tier 2: CoFe4Nd
Logged qcmd tier 2: Co2Fe15Nd2
Logged qcmd tier 2: CoFe2Nd
Logged qcmd tier 2: CoFeNd
Logged qcmd tier 2: CoFeNd2
Logged qcmd tier 2: Co2FeNd
(venv) -> my_db_project git:(master) x
```

Screenshot: CLI running the full workflow



A Cheap First Filter

Before any melting: three composition-only numbers, computed instantly, no lab work required.



VEC

Valence Electron Concentration

Predicts likely crystal structure — solid solution type (BCC / FCC)



δ (delta)

Atomic size mismatch

Small δ favors a solid solution; large δ favors intermetallic phases



ΔH_{mix}

Mixing enthalpy

Predicts compound-forming tendency between elements

Is This Already Known?

Every composition is checked against three independent public materials databases.

Tier 1	Known	Exact match found
Tier 2	Similar	Same elements, different ratio
Tier 3	Partial	Only part of the system is known
Tier 4	Unknown	Nothing found — genuinely novel territory



0.001
eV/atom

Two fully independent databases agreed on a compound's stability within 0.001 eV/atom — strong cross-validation that the result is real.

Databases used: Materials Project • OQMD • Alexandria



A Real Interface

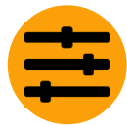
Everything above, in one desktop application.

- Real-time formula validation as you type
- Material classes loaded live from the database
- Automatic sample ID suggestions
- One click: screen, cross-check, calculate, submit

The screenshot shows the Alloy Lab Database Manager application. At the top, a status bar indicates "Sample La1Fe114Si16-20260708-001 added to database". The main interface has a dark theme with various input fields and buttons. A modal dialog box is open, displaying a rocket icon and the message "Sample La1Fe114Si16-20260708-001 added successfully!" with an "OK" button. The main form includes fields for Alloy Formula (La1Fe11.4Si1.6), Unit (at%), Target Mass (g) (10.0), Material Class (Custom...), Excess (e.g., Nd:3) (La:3), and Sample ID (La1Fe114Si16-20260708-002). A "Calculate & Preview" button is visible. Below the form, a "Calculation Results" section displays the following information:

```
=====
Calculation Results
=====
Formula: La1Fe11.4Si1.6 (as at%)
Target mass: 10.0g
Material class: test_class
Sample ID: La1Fe114Si16-20260708-001

Screening Results:
VEC = 7.19
δ = 0.126
ΔH_mix = -2.9 kJ/mol
```



Adjustable Relevance

Not every match found is worth seeing.

- Switch between databases with one click
- A slider sets how similar a match must be to matter — each database remembers its own setting
- All three databases are checked once; switching views is instant, no re-querying

Alloy Lab Database Manager

✓ Calculation complete - ready to submit

[New Entry](#) [Quick Lookup](#) [Summary](#)

Alloy Formula:

✓ Valid: La=7.1, Fe=B4.3, Si=8.6

Unit:

Target Mass (g):

Material Class:

Excess (e.g., Nd₃):

Sample ID: [Auto-generate ID](#)

[Calculate & Preview](#) [Submit to Database](#)

Literature DB: ☒ Materials Project ☐ OQMD ☐ Alexandria Cutoff:

Mass Breakdown:

Element	at%	wt%	target(g)	weigh(g)
Fe	84.29	79.24	7.9243	7.9243
Si	9.57	4.05	0.4053	0.4053
La	7.14	16.70	1.6704	1.7205

Materials Project literature check (cutoff=0.50) -- 3 of 6 shown

Tier 2	LaFeSi4	0.019 eV/atom	(known), distance=0.200
Tier 2	LaFeSi5	0.036 eV/atom	(computed only), distance=0.271
Tier 2	La(FeSi)2	stable	(known), distance=0.443

[Calculate & Preview](#) [Submit to Database](#)

Literature DB: ☐ Materials Project ☒ OQMD ☐ Alexandria Cutoff:

$\Delta H_{mix} = -2.4$ kJ/mol

Mass Breakdown:

Element	at%	wt%	target(g)	weigh(g)
Fe	84.29	79.24	7.9243	7.9243
Si	9.57	4.05	0.4053	0.4053
La	7.14	16.70	1.6704	1.7205

OQMD literature check (cutoff=0.40) -- 2 of 8 shown

Tier 2	Fe9LaSi4	0.018 eV/atom	(known), distance=0.200
Tier 2	Fe2LaSi	0.579 eV/atom	(computed only), distance=0.343

[Calculate & Preview](#) [Submit to Database](#)

Literature DB: ☐ Materials Project ☐ OQMD ☒ Alexandria Cutoff:

Alexandria literature check (cutoff=0.30) -- 11 of 32 shown

Tier 2	Fe10LaSi2	0.063 eV/atom	(computed only), distance=0.074
Tier 2	Fe4LaSi	0.423 eV/atom	(computed only), distance=0.116
Tier 2	Fe9LaSi4	0.015 eV/atom	(computed only), distance=0.200
Tier 2	Fe12LaSi6	0.162 eV/atom	(computed only), distance=0.230
Tier 2	Fe17La2Si9	0.030 eV/atom	(computed only), distance=0.236
Tier 2	Fe1LaSi	0.575 eV/atom	(computed only), distance=0.243
Tier 2	Fe7La3Si2	0.134 eV/atom	(computed only), distance=0.260
Tier 2	Fe4LaSi2	stable	(computed only), distance=0.271
Tier 2	Fe12La2Si7	0.115 eV/atom	(computed only), distance=0.271
Tier 2	Fe8LaSi5	0.033 eV/atom	(computed only), distance=0.271
Tier 2	Fe12La2Si3	0.161 eV/atom	(computed only), distance=0.297

Screenshot: literature database selector & cutoff slider



- Sample counts, class breakdowns, and trends at a glance
- Useful for quick sanity checks between lab sessions
- Shareable with collaborators later on

Screenshot: Metabase dashboard



From Data to Discovery

Real characterization data now flows directly into the database.



XRD

RP1a 101 peaks $a = 15.627 \text{ \AA}$

RP2a 96 peaks $a = 15.630 \text{ \AA}$

RP3a 98 peaks $a = 15.744 \text{ \AA}$



VSM (sample RP1a)

Saturation moment (Ms) **6.883 emu**

Remanence (Mr) **-0.024 emu**

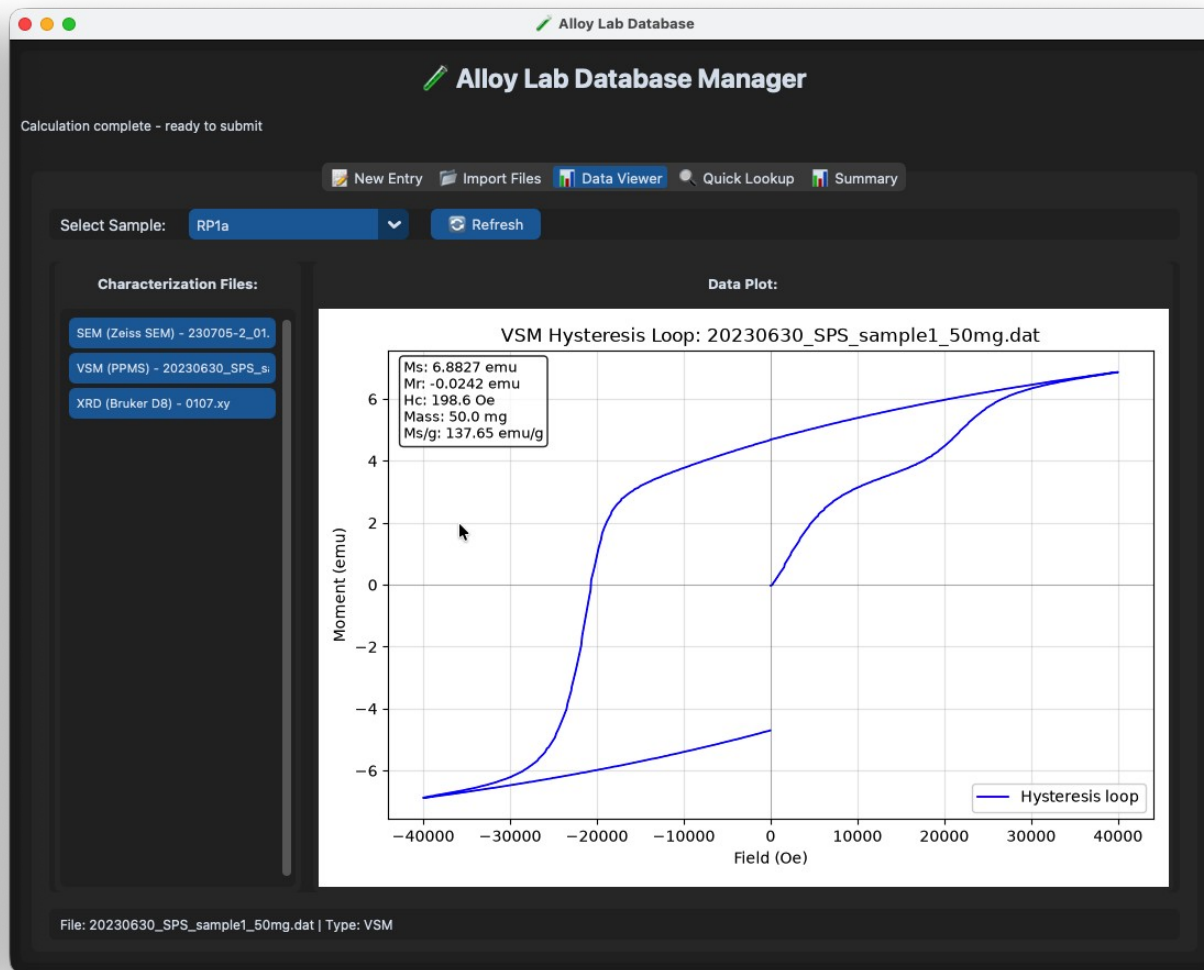
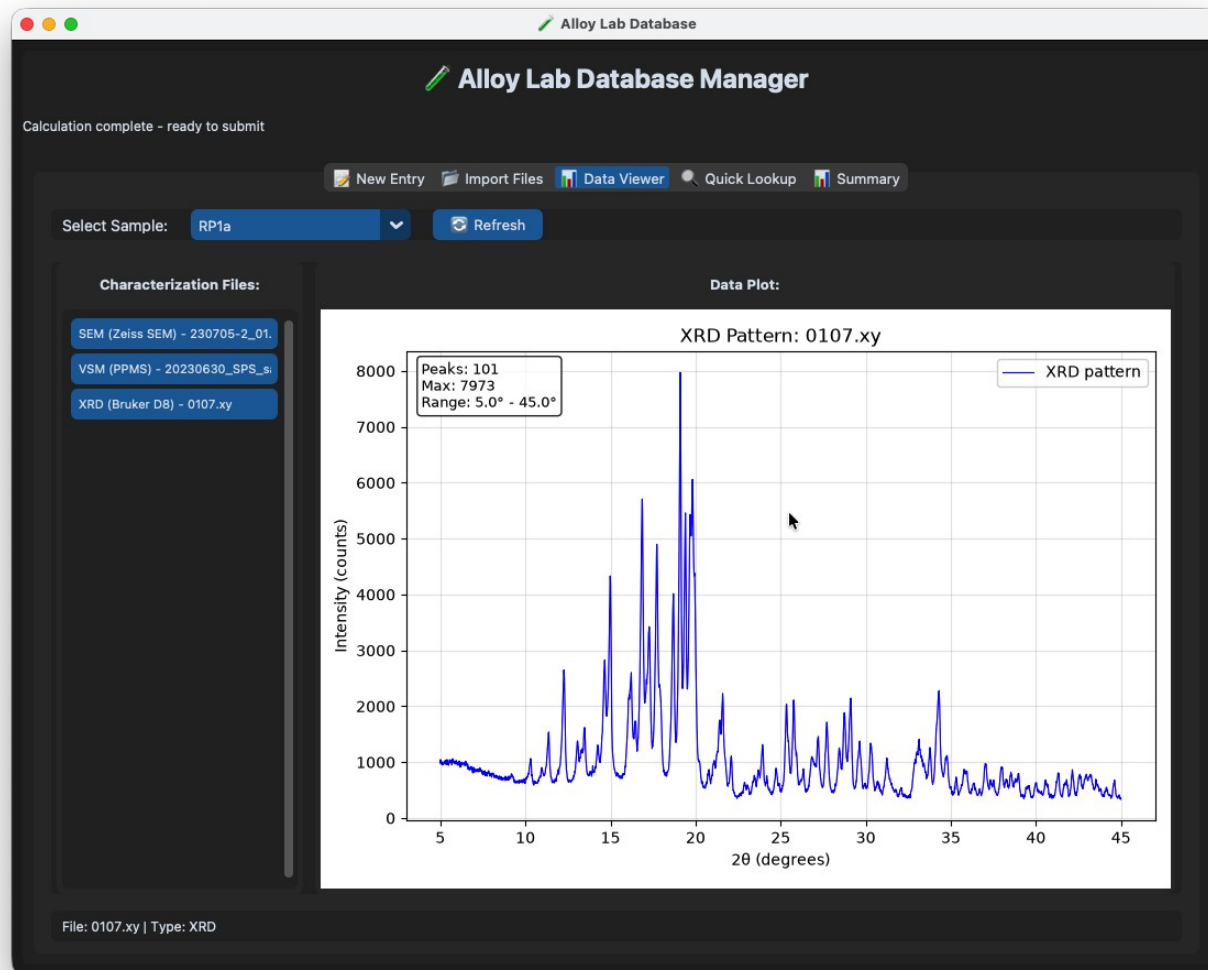
Coercivity (Hc) **198.6 Oe**

Ms per gram **137.65 emu/g**



From Data to Discovery

Real characterization data now flows directly into the database.





SEM — Integrated

The third and final characterization module. Metadata now flows directly into the database, same as XRD and VSM.



Example extracted metadata

Magnification **1.00 KX**

EHT (voltage) **10.00 kV**

Working distance **7.1 mm**

Pixel size **277.3 nm**

Detector **SE2 / InLens**

690

real SEM images imported

Importing one at a time was correct but slow — 5–10 minutes for the full set. Batch import solved the bottleneck without touching the parser itself.



SEM — Integrated

The third and final characterization module. Metadata now flows directly into the database, same as XRD and VSM.



Example extracted metadata

Alloy Lab Database Manager

Summary loaded

New Entry Import Files Data Viewer Quick Lookup Summary

Select Folder: /Users/r/Desktop/NdFeB_data/sorted_v2/sem/test Browse Refresh

Import	File Name	Type	Size	Sample ID
Import	File Name	Type	Size	Sample ID
Import	File Name	Type	Size	Sample ID
Import	File Name	Type	Size	Sample ID
✓	230811-2226_Sample711_20260709_093255.tif	tif	871.6 KB	HT-1000-001
✓	230811-2226_Sample712_20260709_093255.tif	tif	871.6 KB	HT-800-001
✓	230811-2226_Sample713_20260709_093255.tif	tif	871.6 KB	713

Auto-detect Samples Import Selected Select All Deselect All

Manual Entry (for selected file): Enter sample ID manually Apply to Selected (or use dropdown per file below)

Import Log:

```
Import complete: 3 imported, 0 errors
Auto-detected samples for 3 files
Imported: 230811-2226_Sample711_20260709_093255.tif -> 711 (SEM)
SEM: Mag = 1.00 K X, EHT = 0.000 kV
Imported: 230811-2226_Sample712_20260709_093255.tif -> 712 (SEM)
SEM: Mag = 100 X, EHT = 0.000 kV
Imported: 230811-2226_Sample713_20260709_093255.tif -> 713 (SEM)
SEM: Mag = 1.00 K X, EHT = 0.000 kV
Imported: 230811-2226_Sample714_20260709_093255.tif -> 714 (SEM)
SEM: Mag = 5.00 K X, EHT = 0.000 kV
```

Alloy Lab Database Manager

Summary loaded

New Entry Import Files Data Viewer Quick Lookup Summary

Select Sample: NdFeB139-33260715-711 Refresh

Characterization Files:

- SEM (2x4k SEM) - 230811-0126
- SEM (2x4k SEM) - 230811-0126

Data Plot:

SEM Image: 230811-2226_Sample711_20260709_093255.tif

Mag: 5.00 K X
EHT: 0.000 kV
WD: 7.7 mm
Pixel: 237.3 nm



Where Things Stand



Working now

- Core database, credentials & schema
- Mass calculator (pre-alloys, excess %)
- Composition screening (VEC / δ / ΔH_{mix})
- Three literature databases, deduped & cross-checked
- XRD, VSM, and SEM characterization import
- Desktop GUI, five tabs, end to end
- Data sorter, tested on real files



Minor polish (Stage 2)

- Automatic retry for occasional database timeouts
- CLI tool has fallen behind the GUI in features
- VSM plot markers and import progress indicators



What's Next — Stage 2

Stage 2 shifts focus: preparing this data for machine learning, not adding new capture capability.



Element-fraction table

Restructured for fast ML-style
compositional queries



CLI vs. GUI decision

Consolidate around one primary
interface



Curve-fitting & extraction

Deeper features from raw
XRD/VSM curves



Patent records

Composition ranges patented
before publication



Thank You

Built one real problem at a time.

Questions welcome